

Agent Volumes — Pre-Launch Technical Brief

What is Agent Volumes?

Agent Volumes is an open specification for packaging, distributing, verifying, and discovering AI agent components.

The project is designed as a supply-chain and package ecosystem layer for agent systems — analogous to how npm, PyPI, Cargo, and OCI function for conventional software ecosystems.

A versioned package unit is called a **volume**. Registries serving volumes are called **bibliothecas**.

Why This Exists

Current AI agent ecosystems increasingly rely on reusable:

- skills
- tools
- hooks
- commands
- MCP servers
- runtime integrations

However, these components currently lack:

- a unified package identity model
- interoperable dependency semantics
- provenance and integrity guarantees
- standardized advisory workflows
- cross-runtime compatibility metadata
- portable supply-chain tooling compatibility

Agent Volumes attempts to provide this missing package and trust layer.

Core Design Goals

Runtime-neutral packaging

Agent Volumes is intentionally runtime-neutral.

The spec does not assume or privilege a single runtime ecosystem and instead defines portable component abstractions for:

- agents
 - skills
 - tools
 - hooks
 - MCP servers
 - LSP servers
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Supply-chain integrity first

The spec treats trust and verification as first-class concepts.

Core mechanisms include:

- purl-compatible identities
 - immutable content digests
 - normalized file-tree hashing
 - provenance metadata
 - Sigstore/SLSA-aligned workflows
 - advisory metadata
 - trust attachment discovery APIs
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Cross-runtime interoperability

The compatibility model supports:

- Claude Code
- OpenAI Agents SDK
- Codex
- Cursor

- Continue
 - LangGraph
 - MCP-compatible runtimes
 - future runtime profiles
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Reference Implementations

- **shelf** — reference CLI client
- **Alexandria** — reference bibliotheca implementation

These are implementation projects, not governance artifacts.